

Public Chat Portal for AlKhidmat

Kaavish Report
presented to the academic faculty
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1. Introduction

1.1 Problem Statement

Alkhidmat Foundation Pakistan, one of the largest non-profit organizations in the country, handles thousands of daily inquiries from donors, volunteers, and beneficiaries regarding donations, healthcare, and community welfare programs. Currently, these queries are managed manually through phone calls, emails, or in-person visits. This system leads to inefficiencies, delayed responses, redundant questions, and overburdened staff while also alienating overseas or low-literacy users as they struggle to navigate static website content or locate authentic project data.

The lack of an automated communication solution restricts inclusivity and accessibility, especially for marginalized communities or overseas donors seeking quick, verified information. Moreover, Alkhidmat’s operations span multiple domains; healthcare (hospitals, pharmacies, diagnostics, blood banks), education, disaster management, orphan care, women empowerment, and more, creating a vast and complex information network that manual systems cannot efficiently support.

Hence, the problem is twofold:

1. Operational burden due to repetitive, manually handled queries.
2. Limited accessibility for donors and beneficiaries due to language, literacy, and technology barriers.

The motivation behind this project is to bridge this gap by creating an AI-driven, multilingual, multimodal public chat portal that ensures inclusivity, transparency, and efficient information flow which empowers both users and staff.

1.2 Proposed Solution

The proposed solution is an AI-powered Public Chat Portal for Alkhidmat Foundation, a unified platform integrating a Retrieval-Augmented Generation (RAG)-based

LLM with multimodal capabilities (text and image input). The portal will be accessible through Alkhidmat’s official website and WhatsApp Business API to allow seamless communication across all major user groups; donors, patients, blood donors, and beneficiaries.

System Overview:

1. **Frontend Chat Interface:** Accessible through two primary channels:
 - **WhatsApp Interface:** Integrated via the WhatsApp Business API, allowing users to chat directly through WhatsApp using text, voice notes, and image uploads. The interface employs structured message templates and quick-reply buttons for guided workflows, ensuring inclusivity for overseas donors and non-technical users.
 - **Web Interface:** Built in React for responsive and multilingual interaction (English, Urdu, Roman Urdu), providing a modern and intuitive experience for both desktop and mobile users.
2. **Backend Services:** Powered by FastAPI and LangChain, the backend manages query classification, retrieval, and response generation. It includes:
 - **Intent Classification Module:** Determines the user’s domain of inquiry (e.g., Donation, Healthcare, Blood Services, General Support).
 - **Retrieval-Augmented Generation (RAG) Engine:** Combines LLM reasoning with a verified, domain-specific knowledge base to produce contextually accurate and concise responses.
 - **Multimodal Processing Layer:** Handles image understanding and context management for smooth multimodal input handling.
 - **Security and Compliance Layer:** Enforces end-to-end encryption, anonymization of user data, and access control for backend APIs.
3. **Knowledge Base (RAG Datasets):** Curated and periodically updated datasets covering:
 - Donation and payment workflows
 - Healthcare and welfare program details
 - Donor engagement and transparency data
 - CSR and corporate partnership information
 - Accessibility and support resources

4. **Human Escalation Dashboard:** A staff-facing interface where unresolved or sensitive queries are routed to human agents. Staff can respond directly, monitor conversation histories, and update relevant dataset entries.
5. **Analytics and Monitoring Module:** Tracks user engagement, response accuracy, and escalation frequency. Insights from these analytics help refine the system’s knowledge base and improve LLM performance over time.

1.3 Intended User

The chat portal caters to three primary user groups, identified through persona analysis from workshops with Alkhidmat:

1. **Donors (Local, Overseas, Corporate):**

- Seek transparency, convenience, and verified information about donation channels, project impact, and tax compliance.
- Subtypes include recurring donors, new donors, first-time donors, and CSR/corporate donors.

2. **Healthcare Beneficiaries (Patients, Blood Donors, Recipients):**

- Require guidance on healthcare services, test prices, facility locations, doctor availability, ambulance booking, and Zakat-based assistance.
- Includes rural and urban patients with varying digital literacy levels.

3. **General Users (Public Information Seekers):** Represent individuals who approach Alkhidmat for general inquiries such as volunteer opportunities, branch contact details, upcoming drives, disaster relief information, and awareness campaigns. The chatbot provides them with general updates, project overviews, and links to participation or registration forms.

Through intent classification, each user’s domain (donation, healthcare, general inquiry) is automatically detected, ensuring relevant responses and routing. Administrative users (Alkhidmat staff) access a dashboard to manage escalations, update datasets, and broadcast campaigns via WhatsApp APIs.

2. Software Requirement Specification (SRS)

This chapter provides detailed specifications of the system under development.

2.1 Functional Requirements

Functional requirements are grouped into modules corresponding to major system components.

- Core Chatbot Module (Public Facing)
 - Multimodal Input Handling
 - * Text Input: Accept text queries in English, Urdu, and Roman Urdu (basic level).
 - * Image Input: Analyze uploaded images for contextual information (e.g., payment receipts, prescriptions, donation screenshots). Handwritten text recognition is out of scope for MVP.
 - * Voice Input (Phase 2 - Conditional): Convert voice messages to text using Speech-to-Text API. Basic Urdu support only; dialect variations are out of scope. Voice integration depends on successful testing of Phase 1 text-based system.
 - *MVP Scope: Phase 1 focuses on duo-lingual text (English + Urdu) and image input only.*
 - Intent Classification & Domain Identification
 - * Domain Classification: Automatically identify the query domain from four categories:
 - Donation: Payment methods, project details, Zakat/Sadaqah eligibility

- Healthcare: Hospital locations, services, appointments, ambulances
- Blood Services: Donor eligibility, blood bank locations, blood group availability
- General Inquiry: Organization information, volunteering, events, queries about other domains
- *Note: this does not cover the exhaustive list of data for the domains mentioned*
- * Intent Clarification Loop: When queries are ambiguous or confidence score is low, prompt user with clarifying questions before proceeding (e.g., "Are you asking about blood donation or healthcare services?").
- * Confidence Threshold: Queries with confidence score below 0.7 are flagged for human escalation.
- Contextual Response Generation
 - * RAG-Based Retrieval: Retrieve verified information from the domain-structured knowledge base using Retrieval-Augmented Generation.
 - * Persona-Aware Responses: Adapt response length, tone, and detail based on identified user persona (e.g., overseas donor, non-tech-savvy patient, recurring donor).
 - * Follow-Up Context: Maintain session context to handle follow-up questions that depend on prior conversation history.
 - * Session Management: Preserve conversation context throughout the user session to enable multi-turn dialogues.
- Multilingual Response & Accessibility
 - * Language Selection: Allow users to select their preferred language (English or Urdu) at the start of interaction.
 - * Quick Reply Options: Provide button-based shortcuts and pre-defined options for low-literacy users to minimize typing.
 - * Consistent Quality: Ensure response quality is consistent across languages with no discrepancy between English and Urdu outputs.
- Trust & Verification
 - * Branding Display: Show verified Alkhidmat branding, official logos, and chatbot verification messages to assure users of authenticity.
 - * Official Channel Verification: Display verified WhatsApp Business number and official website domain to prevent scam concerns.

- Guardrails & Safety
 - * Content Moderation: Implement guardrails to prevent inappropriate responses or off-topic conversations.
 - * Privacy Protection: Ensure no personal data is retained beyond the session lifetime.
 - * Fallback Responses: Provide predefined safe responses when the system cannot confidently answer a query.
- Information Retrieval & Data Management Module

The knowledge base contains periodically updated, domain-structured information covering (note: this is not an exhaustive list of the data for each domain):

 - RAG Knowledge Base (General Information)

Note: Knowledge base is NOT real-time; it is periodically updated by administrators.

 - * Donation Support:
 - Active and verified campaign details (healthcare, education, orphan care, women empowerment, disaster relief)
 - Payment methods and step-by-step guidance (Easypaisa, JazzCash, bank transfer, international cards, PayPal)
 - Zakat and Sadaqah eligibility criteria for different programs
 - Donation workflow and confirmation processes
 - Transparency reports and impact summaries
 - * Healthcare Information:
 - List of Alkhidmat hospitals, clinics, labs, and pharmacies with addresses and coordinates
 - Service offerings per facility (OPD, diagnostics, pharmacy, ambulance)
 - Operating hours and appointment procedures
 - Consultation fees, test charges, and medicine pricing
 - Doctor availability, specializations, and gender-specific staff information
 - Emergency helpline numbers and ambulance booking procedures
 - * Program Information:
 - Summaries of Alkhidmat’s 12+ domains (education, water, orphan care, community services, disaster management)

- Women-centric initiatives (Bano Qabil, maternity services, women empowerment)
- Youth and student programs
- Corporate Social Responsibility (CSR) partnership opportunities
- * General Information:
 - Organization history, mission, and credibility verification
 - Volunteering opportunities and community event schedules
 - FAQs on trust, transparency, and fund utilization
- Blood Donation Microservice (Operational Data)

A separate microservice containing transactional and inventory information integrates with the main system to handle blood donation queries:

 - * Donor Eligibility Validation: Check eligibility based on age, weight, hemoglobin levels, blood pressure, and donation intervals (WHO and Alkhidmat internal SOPs).
 - * Blood Bank Locations: Retrieve nearest Alkhidmat-licensed blood bank addresses and contact information based on user location.
 - * Blood Group Availability: Query current blood stock levels and urgent blood group requirements.
 - * Donation Process Information: Provide step-by-step donation process details (registration, screening, donation, post-care instructions).
 - * Donor Network Registration: Allow users to register for SMS/WhatsApp alerts when their blood group is urgently needed.
 - * Post-Donation Guidelines: Provide medical instructions (hydration, rest, when to resume activities).
 - * Recipient Support: Guide blood recipients on required documents (CNIC, prescription), cross-match duration, reservation policies, and service charges.
- User Profiling & Persona Management (Contextual Database)

Maintain a separate database to store user information and enable personalized interactions:

 - * Persona Classification: Classify users into personas based on interaction patterns using a donor-beneficiary matrix:
 - Donor personas: Overseas, Local, Recurring, First-Time, Corporate, Non-tech Savvy

- Healthcare personas: Rural/Urban Patient, New Patient, Non-tech Savvy Patient
 - Blood service personas: New/Recurring Blood Donor, Blood Recipient
 - * Interaction History Tracking: Record user query patterns, preferred language, interaction frequency, and topics of interest.
 - * Dynamic Persona Updates: Update user persona classification after each interaction based on new behavioural data.
 - * Context-Aware Response Adaptation: Tailor response length, explanation depth, and recommendations based on identified persona (e.g., provide detailed explanations for first-time donors, concise responses for recurring donors).
 - * Personalized Recommendations: Suggest relevant campaigns, services, or programs based on user history and persona.
- Human Agent Dashboard (Staff Facing)
 - Escalation & Routing
 - * Automatic Escalation: Forward queries with confidence score below 0.7 to relevant department-based chat agents.
 - * Domain-Based Routing: Route escalated queries to specialized agents (Donation team, Healthcare team, Blood Services team, General Support).
 - * Sensitive Query Handling: Automatically escalate queries involving complaints, refunds, or personal grievances to human agents.
 - Conversation Management
 - * Real-Time Takeover: Allow human agents to join ongoing conversations within the same chat thread without disrupting user experience.
 - * 24-Hour WhatsApp Window: Enable agents to respond within the 24-hour WhatsApp Business API interaction window.
 - * Pre-Approved Templates: Provide agents with Meta-approved message templates for initiating conversations outside the 24-hour window.
 - * Conversation History Access: Enable agents to view full conversation context before responding.
 - Knowledge Base & Dialogue Management

- * Content Updates: Allow administrators to add, modify, or archive FAQs and program information in the RAG knowledge base.
 - * Dialogue Flow Definition: Enable admins to define and update pre-approved greeting messages, fallback responses, and guided conversation flows.
 - * Guardrail Configuration: Provide admin interface to set and adjust content moderation rules and response boundaries.
- Role-Based Access Control
 - * Chat Agent Access: Grant domain-specific agents access only to queries within their department.
 - * Admin Privileges: Provide full system access to administrators for knowledge base management, guardrail settings, and user management.
 - * Audit Trail: Log all agent actions (view, respond, resolve) with timestamps for accountability.
- Analytics & Logging Module
 - Usage Tracking
 - * Query Metrics: Record total queries, query volume by domain, peak usage times, and average session duration.
 - * Response Time Monitoring: Track response time per query (target: 3-15 seconds based on complexity).
 - * Escalation Rate Tracking: Monitor percentage of queries escalated to human agents by domain.
 - User Feedback Collection
 - * Post-Interaction Rating: Prompt users for thumbs-up/thumbs-down rating 15-20 minutes after their last query.
 - * Follow-Up Query Check: Send a message asking if the user has any remaining questions before requesting feedback.
 - * Satisfaction Score Calculation: Aggregate user satisfaction metrics over time.
 - Performance Analysis
 - * Answer Accuracy Monitoring: Track percentage of queries resolved by AI without escalation (target: $\geq 90\%$).

- * Domain Performance: Analyze which domains have highest accuracy, longest response times, or most escalations.
 - * Trend Identification: Highlight common query topics and identify emerging information needs that require knowledge base updates.
- Reporting & Insights
 - * Dashboard Visualization: Provide real-time analytics dashboard for administrators showing KPIs and trends.
 - * Export Functionality: Generate CSV/PDF reports of usage metrics for management review.
 - * Anomaly Detection: Flag unusual query patterns (e.g., spam, repeated failures, sudden traffic spikes).
- Out of Scope
 - Handling of regional Urdu dialects is beyond the current scope. The system is designed to support only standard Urdu syntax and semantics.
 - The chatbot does not process handwritten text to extract text from images.
 - Support for additional regional languages such as Punjabi, Sindhi, Pashto, or Balochi is out of scope. The MVP exclusively supports Urdu and English text input.
 - The chatbot does not produce voice-based or audio-generated replies. All communication remains text-based for now.
 - The LLM does not store personal data beyond basic identifiers (name, contact number, email). If such data already exists in the Alkhidmat database, it is retrieved for continuity; otherwise, the LLM treats the user as new. No long-term memory or profiling is maintained.
 - Real-time data access is out of scope. The LLM retrieves information from the existing database, which requires manual updates. If the database is not updated, the system will rely on previously stored information and cannot refresh data automatically.
 - Following features are out of scope as for phase 1 of development; In case of a successful phase 1 completion, these will be added to the scope.
 - * Bilingual speech input support for chatbot interaction, enabling automatic transcription and intent detection from spoken English or Urdu.

- * Enhanced understanding and generation of Roman Urdu text for more natural and localized user communication.
- * Advanced WhatsApp broadcasting and personalized campaign messaging to improve outreach effectiveness.

2.2 Use Cases

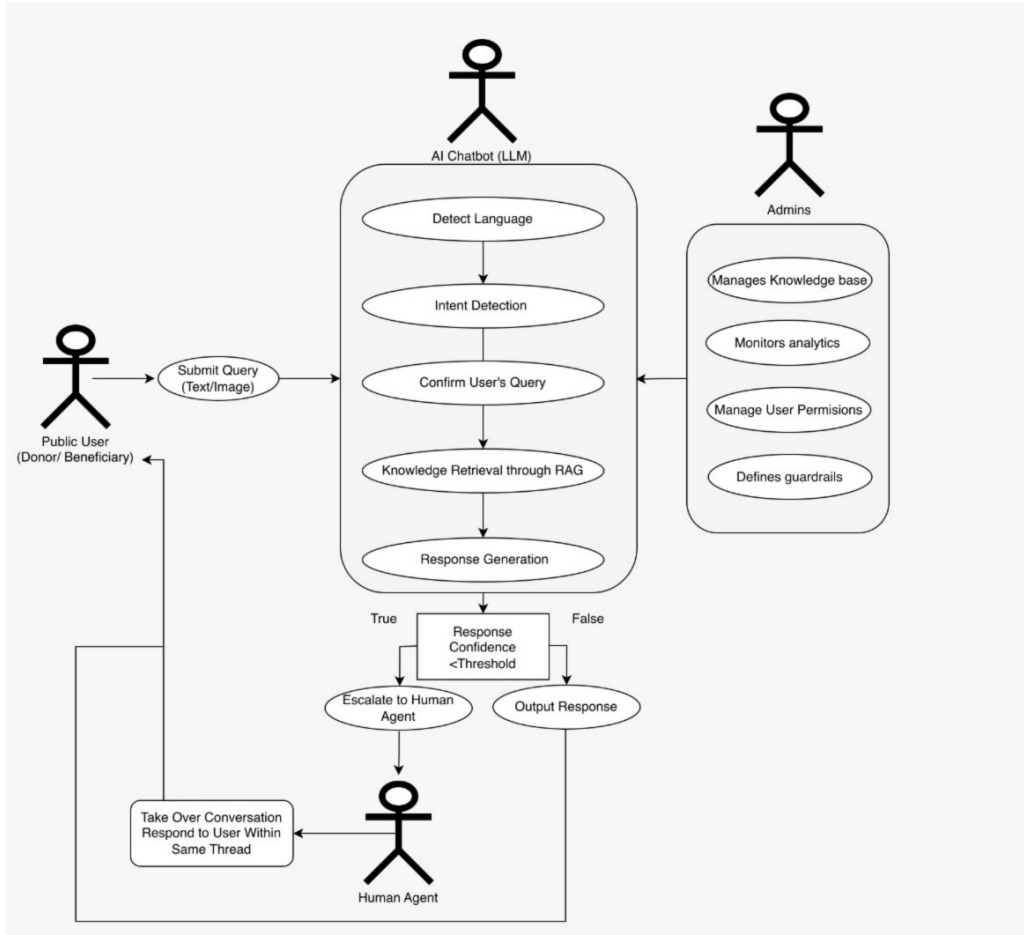


Figure 2.1: Use Case Diagram of the Public Chat Portal for Alkhidmat

2.3 Non-functional Requirements

Table 2.1: Non-Functional Requirements

Category	Description	Example
Performance	The system should respond within 3 to 15 seconds per query, depending on the complexity.	Simple queries (e.g., blood bank locations) should return results in 3 seconds. Complex queries (e.g., eligibility criteria) may take up to 15 seconds.
Accuracy and Relevance	Ensure the system provides $\geq 90\%$ accuracy and relevance in its responses to ensure reliability.	If a user asks for blood donation instructions, the response should be factually correct and relevant in 90% of cases.
Usability / Accessibility	Support multilingual responses with consistent quality in both languages (e.g., English and Urdu). Include clear button-based flows for non-technical users.	The system should provide responses in Urdu with the same clarity and accuracy as in English, without discrepancies.
Reliability	Ensure the system maintains $> 90\%$ answer accuracy and minimal human routing. If accuracy falls below a threshold, route to a human agent.	If a user asks for donation details and the system’s confidence is below 70%, it should route to a human agent.
User Satisfaction (KPI)	Measure user satisfaction by asking for feedback 15–20 minutes after the last query with thumbs up/down ratings.	After a session, if the user does not respond or replies with “no,” send a rating question (thumbs up/down) to evaluate the system’s effectiveness.
Data Integrity & Backup	Ensure data accuracy and periodic backups of important data (e.g., RAG knowledge base). Implement automated weekly backups and data validation.	Perform weekly automated backups of key data and ensure that data validation occurs during data entry to prevent corruption.

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Table 2.1 (continued): Non-Functional Requirements

Category	Description	Example
Security / Compliance	Implement end-to-end encryption for all communications. Ensure no personal data retention beyond session lifetime.	Personal data should not be retained after the session ends. The system should use encrypted communication protocols to secure sensitive information.
Scalability	Support ≥ 500 simultaneous users without performance degradation using FastAPI + Elasticsearch on Docker/Heroku.	The system should scale to support 500 simultaneous users accessing the platform without any degradation in response time.
Maintainability	Use modular code architecture with clear documentation and API version control to ensure the system is easy to maintain and update.	The codebase should be modular with proper documentation to allow for easy updates and troubleshooting.
Compliance and Ethics	Adhere to Pakistan’s data privacy guidelines and AI ethics principles for NGO use.	Ensure that the system follows local data privacy regulations and complies with ethical AI practices for non-profit organizations.